TOPIC 10

Hypothesis Testing

Deciding between two claims with data: hypotheses, test statistics, p-values, errors, power and the link to confidence intervals

WHAT YOU WILL LEARN

- State null and alternative hypotheses correctly, one- and two-sided
- Carry out a z-test and a one-sample t-test step by step
- Define and interpret the p-value and the significance level α
- Explain Type I and Type II errors, and the power of a test
- Connect tests to confidence intervals
- Avoid the standard misinterpretations of p-values

1. The logic: a courtroom analogy

A confidence interval estimates *how big* something is. A **hypothesis test** answers a yes/no question: *is the evidence strong enough to doubt a claim?* The logic mirrors a trial.

Courtroom	Hypothesis test
The defendant is presumed innocent	The **null hypothesis H_0** is assumed true (nothing special is happening)
Prosecutor must present evidence	We collect data and compute a **test statistic**
Evidence 'beyond reasonable doubt'?	Is the result unlikely if H_0 were true? (**p-value** below **α**)
Verdict: guilty or not guilty	**Reject H_0** or **fail to reject H_0**
'Not guilty' is not 'proven innocent'	Failing to reject does **not** prove H_0 true

2. The six steps

1. State the **hypotheses** H_0 and H_1.
2. Choose the **significance level α** (commonly 0.05) before looking at the data.
3. Check the **assumptions** (random sample, independence, normality or large n).
4. Compute the **test statistic** from the sample.
5. Find the **p-value** (or compare with the **critical value**).
6. **Decide** and state the conclusion in the language of the problem.

3. Hypotheses

	Null hypothesis H_0	Alternative hypothesis H_1 (or H_a)
Role	The default claim, a specific value (always contains =)	What we suspect or want to show
Example (mean)	$\mu = 500$	$\mu \neq 500$ (two-sided), $\mu < 500$ or $\mu > 500$ (one-sided)

Type	H_1	Use when
Two-tailed	$\mu \neq \mu_0$	You care about a difference in **either** direction
Right-tailed	$\mu > \mu_0$	Only an increase matters
Left-tailed	$\mu < \mu_0$	Only a decrease matters

COMMON MISTAKE **Choose the direction before seeing the data**

Switching to a one-sided test after seeing which way the data points halves the p-value unfairly. When in doubt use a two-sided test.

4. The test statistic

A test statistic measures **how far the sample result is from the claim in H_0, in units of standard error** (Topic 07). Big values are surprising if H_0 is true.

$$\text{test statistic} = \frac{\text{estimate} - \text{hypothesised value}}{\text{SE}}$$

Situation	Statistic	Reference distribution
Mean, σ known	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	Standard normal
Mean, σ unknown	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$	t with $n - 1$ df
Proportion	$z = (\hat{p} - p_0)/\sqrt{p_0(1 - p_0)/n}$	Standard normal

For a proportion test the SE uses the **hypothesised** p_0, because we calculate under the assumption that H_0 is true.

5. The p-value and the decision

DEFINITION **p-value**

The probability, **assuming H_0 is true**, of getting a result at least as extreme as the one observed. A small p-value means the data are hard to explain by chance alone.

KEY IDEA **Decision rule**

- If **p-value ≤ α**: reject H_0. The result is **statistically significant**.
- If **p-value > α**: fail to reject H_0. The data do not provide enough evidence against it.

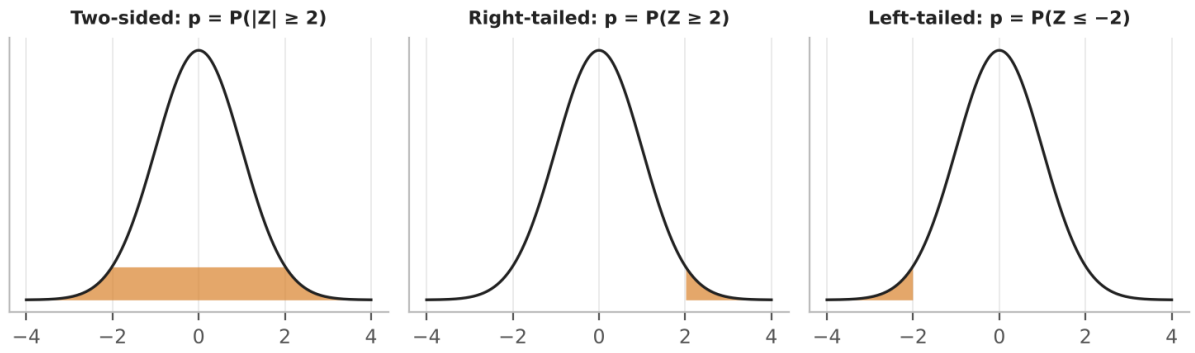


Figure 1. The p-value is the shaded tail area beyond the observed statistic. For z = 2 it is 0.0455 two-sided and 0.0228 one-sided.

Critical-value approach

Equivalent method: find the **critical value** that cuts off α in the tail(s). If the test statistic falls in the **rejection region** beyond it, reject H_0. Two-sided $\alpha = 0.05$ uses $z = \pm 1.96$; one-sided uses 1.645 (Topic 04).

6. Worked example 1: z-test for a mean

> **WORKED EXAMPLE** **Cereal boxes**
> A factory claims boxes average $\mu = 500$ g, with known $\sigma = 30$ g. An inspector weighs $n = 36$ boxes and finds $\bar{x} = 490$ g. Is the claim doubtful? ($\alpha = 0.05$, two-sided)

1. H_0: $\mu = 500$ H_1: $\mu \neq 500$.
2. $\alpha = 0.05$. Assumptions: random sample, σ known, $n = 36 \geq 30$ (CLT).
3. $SE = 30/\sqrt{36} = 5$.
4. $z = (490 - 500)/5 = \mathbf{-2.00}$.
5. Two-sided p-value $= 2 \times P(Z \leq -2) = \mathbf{0.0455}$.
6. Since $0.0455 < 0.05$, **reject H_0**: there is statistically significant evidence the mean differs from 500 g. At $\alpha = 0.01$ we would **fail** to reject, because $0.0455 > 0.01$. The decision depends on α, which is why α is fixed beforehand.

Confidence-interval check: $490 \pm 1.96 \times 5 = [480.2, 499.8]$, which excludes 500, agreeing with the test.

7. Worked example 2: one-sample t-test

> **WORKED EXAMPLE** **Study hours**
> A course claims students study 50 minutes a day on average. A random sample of $n = 25$ students has $\bar{x} = 52.3$ and $s = 8$ (σ unknown). Test at $\alpha = 0.05$, two-sided.

1. H_0: $\mu = 50$ H_1: $\mu \neq 50$.
2. $SE = 8/\sqrt{25} = 1.6$.
3. $t = (52.3 - 50)/1.6 = \mathbf{1.4375}$, $df = 24$.
4. Critical value $t^*(24) = \pm 2.064$. Since $|t| = 1.437 < 2.064$, it is **not** in the rejection region.
5. p-value = **0.1635**, larger than 0.05.
6. **Fail to reject H_0.** The sample mean is higher than 50, but a gap this size is quite plausible by chance with $n = 25$. It is **not** evidence that $\mu = 50$, only a lack of evidence against it.

8. Worked example 3: a proportion

> **WORKED EXAMPLE** **Website conversion**
> The old page converts 40% of visitors. A new page is shown to $n = 200$ visitors and 92 convert ($\hat{p} = 0.46$). Is the new page better? H_0: $p = 0.40$, H_1: $p > 0.40$ (right-tailed, $\alpha = 0.05$).

- Conditions: $np_0 = 80$ and $n(1 - p_0) = 120$, both at least 10.
- SE under $H_0 = \sqrt{(0.4 \times 0.6/200)} = 0.0346$.
- $z = (0.46 - 0.40)/0.0346 = \mathbf{1.732}$.

- One-sided p-value = $P(Z \geq 1.732) = \mathbf{0.0416} < 0.05$, so reject H_0. (A two-sided test would give $p = 0.0833$, which is *not* significant at 0.05. This shows why the direction must be chosen in advance.)

9. Errors and power

Because the decision rests on a sample, it can be wrong in two ways:

	H_0 is actually true	H_0 is actually false
Reject H_0	**Type I error** (false alarm), probability α	Correct: probability = **power** $(1 - \beta)$
Fail to reject H_0	Correct, probability $1 - \alpha$	**Type II error** (missed effect), probability β

- **α = P(Type I)**: set by us. Choosing $\alpha = 0.05$ means a true null is wrongly rejected about 5% of the time.
- **β = P(Type II)**: depends on the true effect size, σ, n and α.
- **Power = 1 − β**: the chance of detecting a real effect. 80% is a common target.

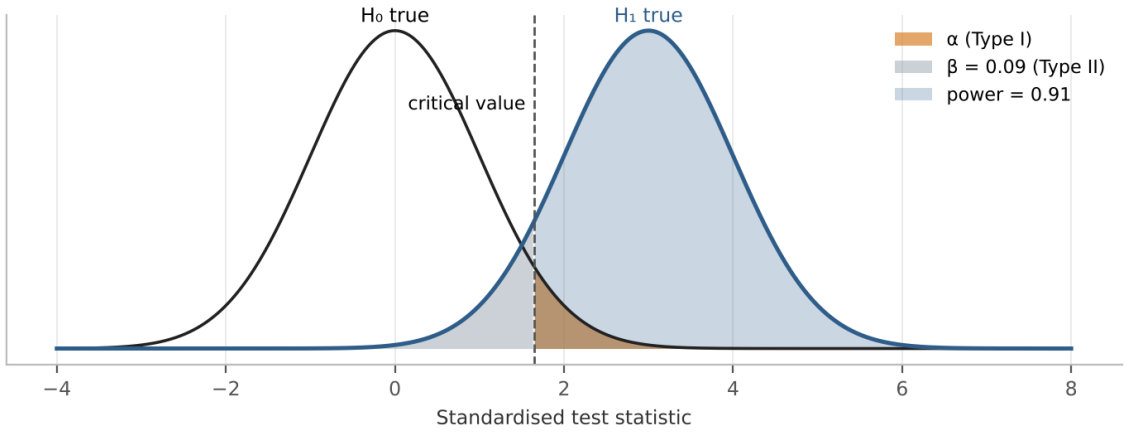


Figure 2. Two sampling distributions: H_0 (black) and one possible truth under H_1 (blue). Moving the cut-off right lowers α but raises β.

What increases power?

Change	Effect on power	Cost / note
Larger n	↑ greatly	Costs more data
Larger true effect	↑	Not under your control
Smaller σ (cleaner data)	↑	Better design or measurement
Larger α	↑	More Type I errors
One-sided instead of two-sided	↑	Only if direction is justified beforehand

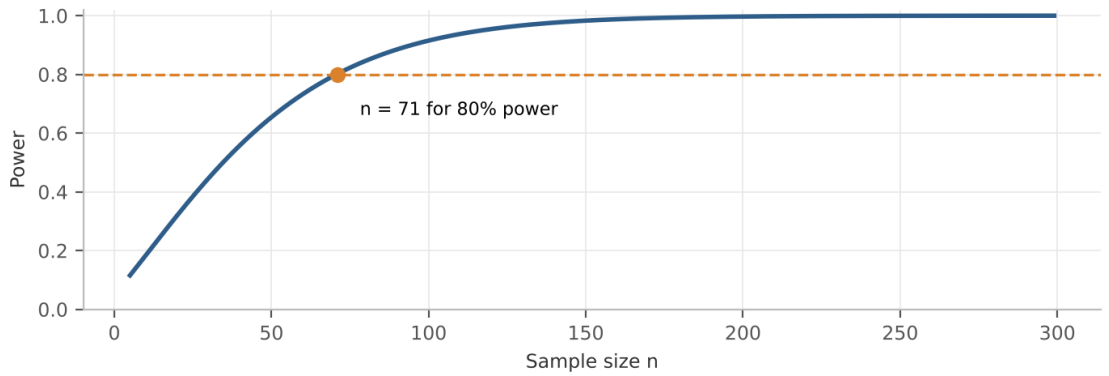


Figure 3. Power of a two-sided z-test (α = 0.05) to detect a true mean of 105 when H_0 says 100 (σ = 15). About n = 71 gives 80% power; with n = 36 power is only 0.52.

KEY IDEA **Why 'fail to reject' is not 'accept'**

A non-significant result can mean H_0 is true **or** that the test lacked power (small n, big σ). Absence of evidence is not evidence of absence, especially for small samples.

10. Tests and confidence intervals agree

A two-sided test at level α rejects H_0: μ = μ_0 **exactly when** μ_0 lies outside the (1 − α) confidence interval. The interval is therefore a richer answer: it tells you **which** hypothesised values are ruled out and how large the effect might be.

	Hypothesis test	Confidence interval
Answers	Is there evidence against μ_0?	What values of μ are plausible?
Gives	p-value and a decision	Range of plausible effect sizes
Shows size of effect?	No	Yes

11. Misinterpretations of p-values

Wrong	Right
'p = 0.03 means there is a 3% chance H_0 is true.'	p is computed **assuming** H_0 is true; it is not the probability of H_0.
'p = 0.03 means the result is 97% likely to be real.'	No. It says data this extreme are rare under H_0.
'p > 0.05 proves no effect.'	It means insufficient evidence; maybe low power.
'A smaller p means a bigger effect.'	p mixes effect size **and** sample size. A tiny, trivial effect can be 'significant' with huge n.
'p = 0.049 and p = 0.051 are completely different.'	The 0.05 line is a convention; report the actual p and the effect size.

Statistical versus practical significance

With n = 1,000,000 a difference of 0.01 points can be 'highly significant' yet meaningless. Always report an **effect size** and a **confidence interval**, and ask whether the difference matters in practice.

Multiple testing and p-hacking

If you run 20 independent tests at $\alpha = 0.05$ with every null true, the chance of at least one false positive is $1 - 0.95^{20} = 0.64$. Testing many things and reporting only the significant ones ('p-hacking') inflates false discoveries. Fix the hypotheses in advance, or adjust: the **Bonferroni** correction uses α/k for k tests.

12. In Python

```
import numpy as np
from scipy import stats

# one-sample t-test from raw data
data = np.array([55, 48, 60, 52, 47, 58, 51, 49, 56, 53])
print(stats.ttest_1samp(data, popmean=50))

# z-test for a mean (sigma known), by hand
xbar, mu0, sigma, n = 490, 500, 30, 36
z = (xbar - mu0) / (sigma / np.sqrt(n))
print(z, 2 * stats.norm.sf(abs(z)))               # -2.0, 0.0455

# proportion test (exact binomial, newer scipy)
print(stats.binomtest(92, 200, p=0.40, alternative="greater").pvalue)
```

SUMMARY

- H_0 is the default claim; H_1 is the alternative. We test H_0 and either reject it or fail to reject it.
- Test statistic = (estimate − hypothesised value)/SE; p-value = probability of results at least this extreme if H_0 is true.
- Reject H_0 when $p \le \alpha$. α is the Type I error rate; β the Type II rate; power $= 1 - \beta$.
- Power rises with n, effect size, α and cleaner data.
- A two-sided test and a confidence interval give the same decision.
- A p-value is not the probability that H_0 is true, and 'not significant' is not 'no effect'.

Practice questions

Try each question before reading the answer underneath it.

Q1. Write H_0 and H_1 for: 'A battery maker claims at least 20 hours of life; we suspect less.'

Answer. H_0: $\mu \ge 20$ (or $\mu = 20$); H_1: $\mu < 20$ (left-tailed).

Q2. $z = -2.5$ in a two-sided test. Find the p-value and decide at $\alpha = 0.05$.

Answer. $p = 2 \times P(Z \le -2.5) = 0.0124 < 0.05$, reject H_0.

Q3. A p-value is 0.20. A manager says 'so the new method has no effect.' Respond.

Answer. Not justified: $p = 0.20$ only means the data are not strong evidence against H_0. The test may have low power, so a real effect could remain undetected.

Q4. What is a Type I error in a medical test for a disease?

Answer. Concluding a healthy person has the disease (false positive).

Q5. If α is lowered from 0.05 to 0.01 with n fixed, what happens to power?

Answer. Power decreases (β increases), because it becomes harder to reject H_0.

Q6. A 95% CI for μ is [12.4, 15.8]. Would a two-sided test of H_0: $\mu = 16$ at $\alpha = 0.05$ reject?

Answer. Yes. 16 lies outside the 95% interval, so H_0 is rejected.